

# Detailed proof of the minimax lower bound on the orbit space, in the $\phi$ -chart

(validation note for the orbital–MDL plan; multiplicative regime)

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(working note)

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## Abstract

We give a complete proof of the minimax lower bound  $\inf_{\hat{f}} \sup_{f^* \in \Sigma_{\text{inv}}(s, L)} \mathbb{E} h^2(P^*, \hat{f}) \gtrsim n^{-2s/(2s+d)}$ ,  $d = \dim(E/G)$ , for estimation of a  $G$ -invariant density, working in the  $\phi$ -chart (multiplicative/twin regime). The argument has three honest pillars: (a) a *reduction* showing the orbit projection is sufficient, so the problem is exactly density estimation on the  $d$ -dimensional quotient; (b) a Varshamov–Gilbert hypothesis cube of  $G$ -invariant bumps with controlled squared-Hellinger separation and Kullback–Leibler radius; (c) Fano’s inequality and a bias/complexity balance. The role of the  $\phi$ -chart is shown to be *transparent*: since squared Hellinger is invariant under the change of variables, the bound transfers verbatim, and the heavy tails of the multiplicative regime are irrelevant to the lower bound because the perturbations are compactly supported. This validates the lower-bound side of the targeted contribution.

## 1 Setting and the reduction to the quotient

We use the notation of the orbital plan.  $E$  carries a compact group  $G$  acting by isometries (in the  $\phi$ -chart  $\phi(E)$ );  $\pi : E \rightarrow E/G$  is the quotient map,  $\mu$  a  $G$ -invariant reference measure,  $\mu_G = \pi_*\mu$ . Write  $M := E/G$  with the infimum orbital metric  $d_G$  and  $d := \dim M = \dim E - \dim(\text{orbit})$ . A  $G$ -invariant density  $f^* = \tilde{f}^* \circ \pi$  corresponds to a density  $\tilde{f}^*$  on  $(M, \mu_G)$ . The squared Hellinger distance is  $h^2(P, Q) = \frac{1}{2} \int (\sqrt{p} - \sqrt{q})^2 d\mu$ , and  $H := \sqrt{h^2}$  is the Hellinger metric.

**Lemma 1.1** (Sufficiency of the orbit projection). *Let  $\mathcal{F}_{\text{inv}}$  be a family of  $G$ -invariant densities on  $E$  and  $X_1, \dots, X_n \stackrel{iid}{\sim} P_f$ ,  $f \in \mathcal{F}_{\text{inv}}$ . Then  $(\pi(X_1), \dots, \pi(X_n))$  is sufficient for  $f$ , are i.i.d. from  $\tilde{P}_{\tilde{f}} = \pi_* P_f$  on  $(M, \mu_G)$ , and  $h_E^2(P_f, P_g) = h_M^2(\tilde{P}_{\tilde{f}}, \tilde{P}_{\tilde{g}})$ . Consequently any estimator  $\hat{f}(X_1, \dots, X_n)$  can be replaced, without increasing the risk, by one depending only on  $(\pi(X_i))_i$ , and*

$$\inf_{\hat{f}} \sup_{f \in \mathcal{F}_{\text{inv}}} \mathbb{E} h_E^2(P_f, \hat{f}) = \inf_{\tilde{f}} \sup_{\tilde{f} \in \tilde{\mathcal{F}}} \mathbb{E} h_M^2(\tilde{P}_{\tilde{f}}, \hat{\tilde{f}}).$$

*Proof.* Invariance  $f = \tilde{f} \circ \pi$  gives  $f(X_i) = \tilde{f}(\pi(X_i))$ , so the likelihood factors through  $\pi(X_i)$ : sufficiency, and the pushforward of an i.i.d. sample is i.i.d. For the Hellinger identity, invariant densities make the affinity integrate to the quotient,  $\int_E \sqrt{fg} d\mu = \int_M \sqrt{\tilde{f}\tilde{g}} d\mu_G$ . For the equality of minimax risks (which is all the lower bound needs), note that under any  $P_f$  with  $f$  invariant, the conditional law of  $X_i$  given  $\pi(X_i) = o$  is the (normalised)  $G$ -invariant measure on the orbit  $\pi^{-1}(o)$ , which does *not* depend on  $f$ : the within-orbit position is *ancillary* with  $f$ -free law. Hence the

experiment  $(X_{1:n}; \{P_f\})$  is Le Cam-equivalent to  $(\pi(X_{1:n}); \{\tilde{P}_{\tilde{f}}\})$  augmented by independent  $f$ -free randomisation, and the two minimax risks coincide; squared Hellinger, being invariant, is the same loss on both sides.  $\square$

Lemma 1.1 reduces the lower bound to density estimation of  $\tilde{f}^*$  on the  $d$ -dimensional space  $(M, \mu_G)$  from  $n$  i.i.d. samples. The  $\phi$ -chart enters only through the definition of smoothness (below); the estimation problem itself is the standard one on  $M$ .

## 2 Standing regularity

**Assumption 1** (Regular chart on the quotient). There is a coordinate ball  $U \subseteq M$  on which  $(M, d_G, \mu_G)$  is Ahlfors  $d$ -regular:  $c_M^{-1} r^d \leq \mu_G(B(x, r)) \leq c_M r^d$  for  $x \in U$ ,  $0 < r \leq r_0$ . Equivalently  $U$  is bi-Lipschitz to a Euclidean ball in  $\mathbb{R}^d$  with  $\mu_G$  comparable to Lebesgue measure. (Holds when the  $G$ -action is free with smooth quotient; at non-free strata,  $d$  is the dimension of the top stratum and  $U$  is taken inside it.)

We identify  $U$  with a Euclidean cube  $(0, 1)^d$  and  $\mu_G$  with Lebesgue measure up to the constant  $c_M$ ; this only affects constants below.

**Assumption 2** (Hölder class in the  $\phi$ -chart).  $\Sigma(s, L) = \{\tilde{f} : \|\tilde{f}\|_{\mathcal{C}^s(M)} \leq L\}$ , the Hölder ball of order  $s > 0$ , smoothness measured in the  $\phi$ -chart, and  $\Sigma_{\text{inv}}(s, L) = \{\tilde{f} \circ \pi : \tilde{f} \in \Sigma(s, L)\}$ . Fix a reference density  $f_0 \in \Sigma(s, L/2)$  with  $0 < c_0 \leq f_0 \leq C_1 < \infty$  on  $U$ .

## 3 The hypothesis cube ( $G$ -invariant bumps)

Fix once and for all a  $\mathcal{C}^\infty$  function  $\psi : \mathbb{R}^d \rightarrow \mathbb{R}$  with  $\text{supp } \psi \subseteq (0, 1)^d$ ,  $\int \psi = 0$ ,  $\|\psi\|_\infty \leq 1$ ,  $\|\psi\|_{\mathcal{C}^s} \leq 1$ , and  $\|\psi\|_{L^2} > 0$ .

For a bandwidth  $h = 1/m$  ( $m \in \mathbb{N}$ ), partition  $U$  into  $N \asymp c_M h^{-d}$  disjoint sub-cubes  $\{U_j\}_{j=1}^N$  of side  $h$  and centres  $x_j$ . Define the rescaled bumps and the hypothesis family

$$\psi_j(x) := L h^s \psi\left(\frac{x - x_j}{h}\right), \quad \tilde{f}_\omega := f_0 + \sum_{j=1}^N \omega_j \psi_j, \quad \omega \in \{0, 1\}^N. \quad (1)$$

**Lemma 3.1** (Admissibility). *For  $h \leq h_0$  (so that  $Lh^s \leq c_0/2$ ), each  $\tilde{f}_\omega$  is a probability density on  $M$  and  $\tilde{f}_\omega \in \Sigma(s, c_1 L)$  for a constant  $c_1$  depending only on  $\psi$ .*

*Proof.*  $\int \psi_j = Lh^s h^d \int \psi = 0$ , so  $\int \tilde{f}_\omega = \int f_0 = 1$ ; and  $|\sum_j \omega_j \psi_j| \leq Lh^s \|\psi\|_\infty \leq c_0/2$ , so  $\tilde{f}_\omega \geq c_0/2 > 0$ . The bumps have disjoint supports, and  $\|\psi_j\|_{\mathcal{C}^s} = Lh^s \cdot h^{-s} \|\psi\|_{\mathcal{C}^s} \leq L$  (the  $s$ -th difference quotient of a width- $h$ , height- $Lh^s$  bump is  $O(L)$ ); for disjoint supports the global  $\mathcal{C}^s$  seminorm of the sum equals the maximum over  $j$ , so  $\|\tilde{f}_\omega\|_{\mathcal{C}^s} \leq \|f_0\|_{\mathcal{C}^s} + L \leq c_1 L$ .  $\square$

**Lemma 3.2** (Varshamov–Gilbert [1, Lem. 2.9]). *For  $N \geq 8$  there is  $\Omega \subseteq \{0, 1\}^N$  with  $|\Omega| \geq 2^{N/8}$  and  $\text{Ham}(\omega, \omega') \geq N/8$  for all distinct  $\omega, \omega' \in \Omega$ . Fix such an  $\Omega$  and a base point  $\omega^{(0)} \in \Omega$ .*

## 4 Separation and information radius

**Lemma 4.1** (Squared-Hellinger separation). *For distinct  $\omega, \omega' \in \Omega$ ,*

$$h^2(\tilde{f}_\omega, \tilde{f}_{\omega'}) \geq \frac{1}{8C_1} \text{Ham}(\omega, \omega') L^2 h^{2s+d} \|\psi\|_{L^2}^2 \geq c_2 L^2 h^{2s}, \quad c_2 := \frac{c_M \|\psi\|_{L^2}^2}{64 C_1}.$$

*Proof.*  $(\sqrt{a}-\sqrt{b})^2 = (a-b)^2/(\sqrt{a}+\sqrt{b})^2 \geq (a-b)^2/(4C_1)$  since  $\tilde{f}_\omega, \tilde{f}_{\omega'} \in [c_0/2, C_1+c_0/2] \subseteq [0, 2C_1]$  for  $h \leq h_0$ . Disjoint supports give  $\int(\tilde{f}_\omega - \tilde{f}_{\omega'})^2 = \sum_{j:\omega_j \neq \omega'_j} \int \psi_j^2 = \text{Ham}(\omega, \omega') L^2 h^{2s} \int \psi(\frac{\cdot - x_j}{h})^2 = \text{Ham}(\omega, \omega') L^2 h^{2s+d} \|\psi\|_{L^2}^2$ . Hence  $h^2 = \frac{1}{2} \int(\sqrt{\cdot} - \sqrt{\cdot})^2 \geq \frac{1}{8C_1} \int(\cdot)^2$ . Finally  $\text{Ham} \geq N/8 \asymp c_M h^{-d}/8$ , so  $h^2 \geq c_2 L^2 h^{2s}$ .  $\square$

**Lemma 4.2** (Kullback–Leibler radius). *For every  $\omega \in \Omega$ , with  $P_\omega^{\otimes n}$  the law of the sample,*

$$\text{KL}(P_\omega^{\otimes n} \| P_{\omega^{(0)}}^{\otimes n}) = n \text{KL}(\tilde{f}_\omega \| \tilde{f}_{\omega^{(0)}}) \leq \frac{2}{c_0} n N L^2 h^{2s+d} \|\psi\|_{L^2}^2 \leq c_3 n L^2 h^{2s}.$$

*Proof.*  $\text{KL}(f\|g) \leq \chi^2(f\|g) = \int(f-g)^2/g \leq \frac{2}{c_0} \int(f-g)^2$ , and  $\int(\tilde{f}_\omega - \tilde{f}_{\omega^{(0)}})^2 = \text{Ham}(\omega, \omega^{(0)}) L^2 h^{2s+d} \|\psi\|_{L^2}^2 \leq N L^2 h^{2s+d} \|\psi\|_{L^2}^2$ . With  $N h^d \asymp c_M$  this is  $\leq c_3 n L^2 h^{2s}$  after the i.i.d. tensorisation  $\text{KL}(P^{\otimes n} \| Q^{\otimes n}) = n \text{KL}(P \| Q)$ .  $\square$

## 5 Fano and the balance

**Lemma 5.1** (Fano method [1, Thm. 2.7]). *Let  $\{P_\theta\}$ ,  $(\Theta, H)$  a semimetric. If  $\theta_0, \dots, \theta_M$  ( $M \geq 2$ ) satisfy  $H(\theta_j, \theta_k) \geq 2\delta$  for  $j \neq k$  and  $\frac{1}{M} \sum_{j=1}^M \text{KL}(P_{\theta_j} \| P_{\theta_0}) \leq \alpha \log M$  with  $0 < \alpha < 1/8$ , then*

$$\inf_{\hat{\theta}} \max_{0 \leq j \leq M} P_{\theta_j}(H(\hat{\theta}, \theta_j) \geq \delta) \geq \frac{\sqrt{M}}{1 + \sqrt{M}} \left(1 - 2\alpha - \sqrt{\frac{2\alpha}{\log M}}\right) \geq c_\star > 0.$$

**Theorem 5.2** (Minimax lower bound on the quotient,  $\phi$ -chart). *Under Assumptions 1–2 there are constants  $c, n_0 > 0$  such that for all  $n \geq n_0$ ,*

$$\inf_{\hat{f}} \sup_{f^\star \in \Sigma_{\text{inv}}(s, L)} \mathbb{E} h^2(P^\star, \hat{f}) \geq c n^{-2s/(2s+d)}.$$

*Proof.* By Lemma 1.1 it suffices to prove the bound on  $(M, \mu_G)$ . Take the family  $\{\tilde{f}_\omega : \omega \in \Omega\}$  of Lemmas 3.1–3.2; all lie in  $\Sigma(s, c_1 L)$ , hence (after relabelling  $L \leftarrow L/c_1$ ) in  $\Sigma(s, L)$ .

*Choice of  $h$ .* Put  $h = h_n$  with

$$n L^2 h^{2s+d} = c_4, \quad \text{i.e.} \quad h_n = \left(\frac{c_4}{n L^2}\right)^{1/(2s+d)}, \quad (2)$$

where  $c_4$  is chosen below; then  $N \asymp c_M h_n^{-d} \rightarrow \infty$ , so  $h_n \leq h_0$  and  $N \geq 8$  for  $n \geq n_0$ .

*Separation.* By Lemma 4.1, with  $H = \sqrt{h^2}$ ,  $H(\tilde{f}_\omega, \tilde{f}_{\omega'}) \geq \sqrt{c_2} L h_n^s =: 2\delta_n$ , i.e.  $\delta_n^2 = \frac{c_2}{4} L^2 h_n^{2s}$ .

*Information.* By Lemma 4.2,  $\frac{1}{|\Omega|} \sum_\omega \text{KL}(P_\omega^{\otimes n} \| P_{\omega^{(0)}}^{\otimes n}) \leq c_3 n L^2 h_n^{2s} = c_3 c_4 h_n^{-d}$  using (2). Since  $\log |\Omega| \geq \frac{N}{8} \log 2 \geq c_5 h_n^{-d}$ , the Fano ratio is

$$\frac{\frac{1}{|\Omega|} \sum_\omega \text{KL}}{\log |\Omega|} \leq \frac{c_3 c_4}{c_5} =: \alpha.$$

Choose  $c_4$  small enough that  $\alpha < 1/8$ .

*Fano.* Lemma 5.1 gives  $\inf_{\hat{f}} \max_\omega P_\omega(H(\hat{f}, \tilde{f}_\omega) \geq \delta_n) \geq c_\star$ . By Markov,

$$\inf_{\hat{f}} \sup_{\tilde{f} \in \Sigma(s, L)} \mathbb{E} h^2(\hat{f}, \tilde{f}) \geq \delta_n^2 c_\star = \frac{c_2 c_\star}{4} L^2 h_n^{2s}.$$

Finally, by (2),  $h_n^{2s} = (c_4/(n L^2))^{2s/(2s+d)}$ , so  $L^2 h_n^{2s} = c_4^{2s/(2s+d)} L^{2d/(2s+d)} n^{-2s/(2s+d)}$ , whence the claimed  $c n^{-2s/(2s+d)}$  (the  $L$ -dependence is absorbed into  $c$ ). Pulling back to  $E$  via Lemma 1.1 completes the proof.  $\square$

## 6 Why the $\phi$ -chart and the heavy tails are harmless here

*Remark 6.1* (Chart transparency). The construction lives on the quotient  $M$  in the  $\phi$ -chart, where smoothness is defined (Assumption 2); but the loss  $h^2$  is a change-of-variables invariant, so the separations of Lemma 4.1 and the bound of Theorem 5.2 are identical whether read in the  $\phi$ -chart or pulled back to  $E$  by  $\phi^{-1}$ . No Jacobian or branch issue enters the lower bound:  $\phi$  is fixed and bijective on the relevant compact ball.

*Remark 6.2* (Heavy tails irrelevant to the lower bound). The perturbations  $\psi_j$  are supported in the fixed compact ball  $U$  and the hypotheses share the bounded reference  $f_0$  there; tail behaviour of the multiplicative regime ( $\phi = \log$  on  $(0, \infty)$ , possibly heavy-tailed in the original scale) plays no role. Heavy tails matter only for the *upper* bound, through the affinity/concentration step; the lower bound is a purely local, compactly supported construction. Hence the matching rate  $n^{-2s/(2s+d)}$  is attained on multiplicative data without any tail assumption on the lower-bound side.

*Remark 6.3* (What this validates and what remains). Theorem 5.2 discharges the lower-bound side of the targeted contribution: combined with the orbital-MDL upper bound it yields the exact rate on the quotient, including the multiplicative regime. The two remaining technical points are both on the *upper* side: (i) the twin Jackson estimate on a possibly stratified quotient (orbifold strata), and (ii) the affinity/Hellinger oracle constant under the twin-transported reference measure for heavy-tailed positive data.

## References

- [1] A. B. Tsybakov, *Introduction to Nonparametric Estimation*, Springer, 2009 (Lemma 2.9, Theorem 2.7).
- [2] A. Barron, T. Cover, “Minimum complexity density estimation,” *IEEE Trans. Inform. Theory* 37 (1991) 1034–1054.